

DualStage Enhancing Emotional Connections through Music Visualization for Synchronizing Live and Virtual Performances

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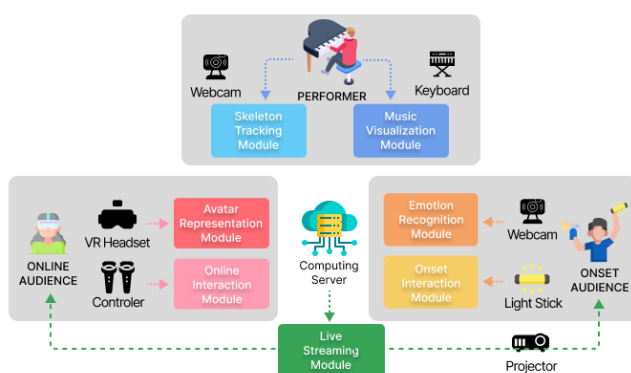
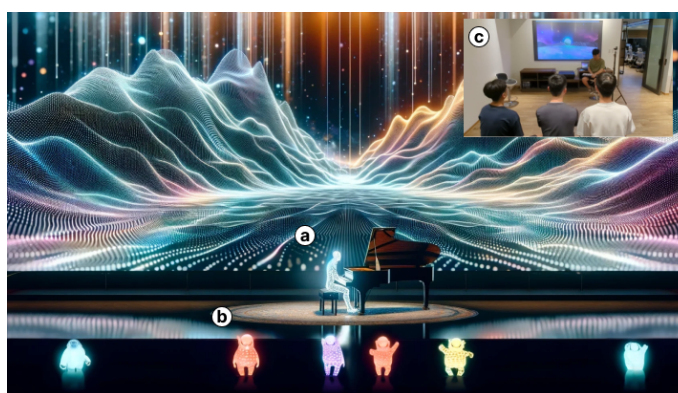


Figure 1: Left: Scenarios of online and onsite audience participating in the concert. (a) The performer is represented as an avatar in the virtual concert. (b) Online audience represented as elves. (c) Onsite audience. Right: Framework of our proposed system.

ABSTRACT

The pandemic has accelerated the adoption of virtual concerts, bridging the physical and virtual worlds for performers and audiences. However, the emotional connection at live concerts remains irreplaceable. To address this issue, we propose DualStage, a real-time emotional visualization system for synchronizing live and virtual performances. The system will visualize the collected data, encompassing various aspects such as the performers' movements, the audience's interactions, and the tempo, rhythm, and intensity of the music. By integrating these elements, the system aims to create a comprehensive and immersive visual experience that enhances the emotional connection between performers and both onsite and online audiences. We conducted an exploratory study with 22 participants to refine the system and enhance cross-reality engagement. The results showed that most participants recognized the feasibility and effectiveness of the system.

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CCS CONCEPTS

• **Human-centered computing** → **Virtual reality; Interaction design.**

KEYWORDS

Multisensory, Live Performance, Music Visualization

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1 INTRODUCTION

During the COVID-19 pandemic, virtual concerts emerged as a new format for many musicians and singers. Virtual platforms such as live streaming and virtual reality became essential tools, providing an alternative way for fans to experience music. However, as the pandemic subsided and restrictions were lifted, music fans began to return actively to live concerts. This shift highlighted the enduring appeal of in-person music events and underscored the importance of understanding the motivations and key factors that drive audience attendance at live concerts. Such insights are crucial

for designing immersive VR experiences. [9] Brown and Knox [3] identified social connectedness as a decisive reason for audience attending live concerts. This sense of community and shared experience is difficult for virtual concerts to replicate fully. Additionally, the uniqueness of live performances is a significant draw for audiences. Attendees relish being part of special events where they can watch artists perform unreleased or rarely played songs and enjoy unique interpretations and improvisations. Lamont [6] highlighted that the strongest musical experiences typically occur during live concerts with others. The physical presence, atmosphere, and energy of a live audience contribute to these intense experiences. In contrast, the primary appeal of virtual concerts lies in their ability to offer unique and visually stunning experiences that are impossible to achieve in a physical setting. Immersive virtual environments can create fantastical visual experiences that enhance the music in new and exciting ways. Vergauwen [12] noted that virtual concerts have the potential to overcome limitations associated with attending live concerts, such as geographical and logistical constraints, suggesting that virtual and live concerts will coexist in the future.

Previous research on virtual concerts has primarily concentrated on enhancing audience engagement and interaction to create a more immersive experience. For instance, the VR concert system developed by Horie et al. [4] generates visual effects based on users' biomedical signals, such as brainwaves and heartbeats, making the audience an integral part of the performance. Similarly, LiVRation by Kasuya et al. [5] allows users to view live performances from multiple angles and control the sound volume independently. Users can customize various audio parameters, such as instrument levels, vocal emphasis, and ambient sound, thereby enhancing their autonomy and creating a more personalized concert experience. Enhancing the sense of presence is also crucial in VR concert experiences. Yakura and Goto [14] explored computational methods to improve co-presence in VR concerts by displaying virtual audience avatars' movements. Yan et al. [15] proposed three interaction modes to boost social presence: shared visual experiences, synchronized activities, and communication through virtual avatars. Beacco et al. [2] focused on creating the illusion of presence by incorporating realistic avatar movements, spatial audio, and interactive environments, making users feel as though they are at a live concert. In addition to VR concerts, the intersection of music and Extended Reality (XR) has expanded, allowing performers and audiences to interact musically with virtual objects, agents, and environments. Turchet et al. [11] investigated this by integrating multisensory feedback and real-time musical interactions in XR environments, enhancing the overall experience. Actualities by Lin et al. [7] integrates live and virtual concerts using multiple displays. Utilizing various sensors, it captures signals from musical instruments and onsite audiences, converting these elements into a virtual world. Visuals are projected onto screens and live-streamed, with interactive components designed for audience engagement, bridging the gap between live and virtual concert experiences.

Unlike previous research, we aim to explore whether visualizing emotions can enhance the social connectedness between online and onsite audiences. To achieve this, our goal is to develop a real-time performance system called DualStage, which can seamlessly synchronize live and virtual performances and visualize all emotion-related elements. The facial expressions and movements of onsite

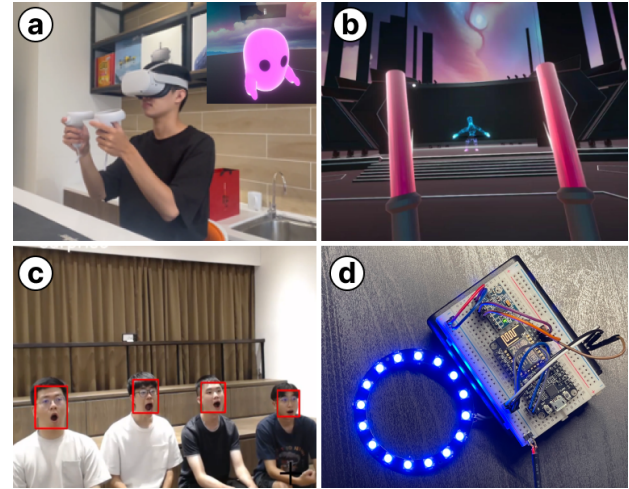


Figure 2: Interaction Mechanism for Online and Onsite audience. (a) Online audience in VR headsets as elf avatars. (b) Online audience controls virtual light sticks with controllers. (c) Real-time detection of onsite audience emotions. (d) Onsite audience interacts with performers using LED sticks.

audiences, the movements of performers, music beats, and the interactions of online audiences are considered emotion-related elements. These elements will be visualized to generate content that will be streamed to screens at both the live concert venue and the virtual concert platform, enhancing the social connectedness between online and onsite audiences.

2 METHOD

DualStage leverages the Unity Engine to create the virtual environment and generate content of emotion-related elements. We integrate Photon to establish a multiplayer network system, allowing content to be streamed to screens at both the live concert venue and the virtual concert platform. The hardware components include a computer, MIDI keyboard, webcams, speakers, projection equipment, light stick devices, and VR headsets. This integration of software and hardware ensures a seamless and immersive virtual concert experience, effectively bridging the gap between physical and virtual audience. Figure 1 illustrates the system framework of the proposed DualStage, which comprises seven main modules. The Performer section includes: (1) Movement Tracking Module and (2) Music Visualization Module. The Online Audience section consists of: (3) Audience Movement and (4) Online Interaction Module. The Onsite Audience section encompasses: (5) Facial Expression Recognition Module and (6) Onsite Interaction Module. Additionally, the system integrates (7) Live Streaming Module, which connects the live stage with the virtual stage.

2.1 Performer

In development stage, we primarily collaborate with pianists. However, as this system can capture music signals via MIDI, it is also compatible with other electronic instrument performers, such as those using electronic drums, electric guitars, or synthesizers.

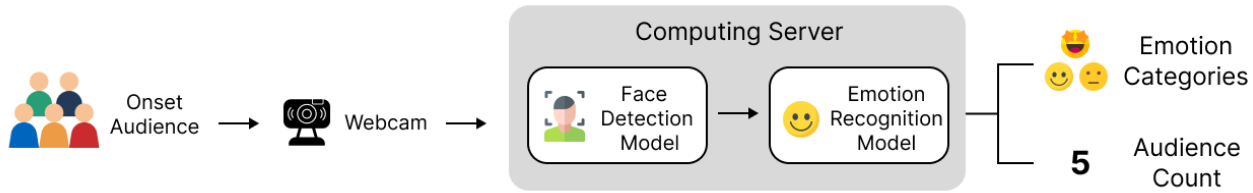


Figure 3: Emotion Recognition Pipeline for Onsite audience

Movement Tracking Module: To accurately capture the performer’s full body posture, we use MediaPipe’s Pose Model [1] to detect the performer’s skeletal structure. The model can detect landmarks of human bodies in images or videos, identify key body locations, analyze posture, and categorize movements. This task outputs body pose landmarks in both image coordinates and 3-dimensional world coordinates. Once MediaPipe captures the performer’s motion data, we utilize Final IK [8] for inverse kinematics (IK) calculations to rig the avatar, ensuring that the virtual avatar moves naturally and accurately mimics the performer’s movements. Based on the IK calculations, we generate particle effects and apply offsets to ensure they align correctly with the performer’s avatar.

Music Visualization Module: We have designed two distinct virtual environments to visualize the music input: a stage scene and a fantasy-style planetary scene for demonstration. We use shaders to create various visual effects for both scenes. Shaders are small programs that run on the GPU to handle rendering effects, allowing for complex and dynamic visual effects in real-time. For the stage scene, we developed visual effects that dynamically respond to the music being played. The input MIDI information influences the visual content and color changes on the stage screens. Specifically, the shaders manipulate the screen textures and colors based on the music’s tempo, rhythm, and intensity, creating an immersive experience for the audience where the visuals are directly tied to the audio. For the planetary scene, the rhythm of the music affects the movement and rotation of the planets. By mapping the beat and tempo of the music to the positional and rotational parameters of the planetary objects, we ensure that the visual effects are synchronized with the audio. Shaders are used here to adjust the lighting, surface textures, and motion blur of the planets, enhancing the fantastical atmosphere and the visual experience more engaging.

2.2 Online Audience

The current system supports the Oculus VR system, enabling immersive virtual concert experiences. It can accommodate up to 100 online audience simultaneously, allowing them to interact with the performance in real-time.

Audience Movement Module Our system enables online audiences to freely move around the virtual concert space. Unlike traditional live concerts where attendees are restricted to a single viewing position, this feature allows audiences to explore the virtual concert from multiple vantage points. Audiences can interact with various elements of the virtual environment and experience the performance from different angles and distances, creating a more dynamic and engaging experience. This freedom fosters a

greater sense of connection to the event and to each other, fostering a sense of community and shared experience even in a virtual setting.

Online Interaction Module We have developed an online interaction mechanism that allows the audience to engage with live performances using VR controllers, as depicted in Figure 2. This module includes two modes: General Interaction Mode and Performance Interaction Mode. In General Interaction Mode, online audiences can wave virtual light sticks in the virtual environment, interacting with other online audience avatars. During the emotional buildup before the chorus or during interludes, online audiences can wave their controllers in sync with the music’s rhythm to trigger blue light orbs. These orbs are sent to the stage, where the system calculates the precise timing and accumulates energy values. When high-energy sections of the song, such as the chorus, arrive, the system releases the accumulated energy, triggering stage fireworks effects. The fireworks vary in size based on the energy released: small (45%), medium (65%), and large (85%). This interaction enhances the immersive experience and creates a dynamic visual spectacle that complements the live performance.

2.3 Onsite Audience

The number of onsite audience primarily depends on the size of the venue. The system itself does not impose any limitations on the number of attendees.

Facial Expression Recognition Module The system captures images of the onsite audience through webcams, which are then processed for face detection and facial expression recognition, as illustrated in Figure 3. Initially, the webcam captures images, and the YOLO model [10] is used for face detection. This model is renowned for its high accuracy and speed, making it suitable for real-time applications by accurately identifying and locating faces in the images. For facial expression recognition, we use a model pretrained on the AffectiveNet dataset. The AffectiveNet dataset [13] provides comprehensive facial expression data, enhancing the model’s ability to recognize a wide range of emotions. This model is based on the MobileNet architecture and has been further fine-tuned with additional data to improve accuracy and adaptability. The detected emotions are aggregated using a voting mechanism to filter out individual variations and identify the predominant emotion, providing a more accurate representation of the overall emotional state of the crowd. The final output includes a count of audiences and categorized emotions, such as happy, neutral, and surprised. These emotions are then visualized in real-time on both the live and virtual stages’ visual effects and lighting dynamics, enhancing the

emotional connection and synchronizing the atmosphere of the virtual concert with the emotions of the onsite audience.

Onsite Interaction Module Inspired by the Smart Light Stick design by Yang et al. [16], we developed an interactive LED stick device using ESP8266, WS2812 LEDs, and MX160 nine-axis accelerometers. This device enables real-time interaction with the stage and supports two modes: General Interaction Mode and Performance Interaction Mode. In General Interaction Mode, the LED sticks held by onsite audience synchronize with the stage lighting. audience can also wave their sticks to launch meteors and light orbs towards the stage, triggering real-time visual effects on the stage screen. During specific performance segments, such as the emotional buildup before the chorus or during interludes, the audience is invited to wave their light sticks in sync with the music. The system records these inputs and accumulates energy values, which are released during peak moments like the chorus, triggering stage fireworks effects.

2.4 Live Streaming Module

We employed Photon to establish a robust system for both onsite and online multi-user virtual concerts. Photon provides the necessary infrastructure for real-time communication and interaction between performers and online audiences. This allows us to seamlessly connect multiple users in a virtual environment, ensuring synchronized experiences and interactions. Additionally, we utilize Unity's Timeline tool to manage performance elements, including lighting, camera movements, and sound effects. The Timeline enables precise choreography of these elements in sync with the music, enhancing the overall visual and auditory experience. To ensure synchronization across all clients, we employ Remote Procedure Calls (RPC) to activate the Timeline simultaneously on all connected devices. This guarantees that all audiences, whether online or onsite, experience the performance in perfect sync. By combining Photon and Unity's Timeline with RPC, our system delivers a highly synchronized and interactive virtual concert experience, bridging the gap between physical and virtual audiences. This integration enhances the synchronization of the performance and strengthens the collective emotional connection of the audiences.

3 RESULTS AND DISCUSSION

3.1 Model Performance

As summarized in Table 1, the fine-tuned model performs well in various cases. We adopted the model with these three emotions as it aligns better with our usage context. This robust performance in emotion recognition highlights its suitability for real-time applications, significantly enhancing the immersive and interactive experience for concert audiences both onsite and online.

3.2 User Study

To evaluate the effectiveness and feasibility of our system, we conducted a user study with 22 participants (15 males and 7 females) aged between 21 and 26 years. The study included questions targeting different modules of the system: the Music Visualization Module, the Onsite Audience Detection Module, and the Online Audience Interaction Module. Each question was rated on a scale from 1 to 5, with 5 being the highest score.

Subset	Num. of Emotions	Emotion (%)
Train	2	91.17
	3 ("Natural","Happy," and "Surprise)	90.92
	5	96.38
	8	99.58
Val	2	84.26
	3 ("Natural","Happy," and "Surprise)	82.55
	5	86.38
	8	86.81
Test	2	82.31
	3 ("Natural","Happy," and "Surprise)	82.31
	5	83.67
	8	85.71

Table 1: Facial expression recognition accuracy after fine-tuning the model.

- Q1: "The deformation of the sphere in response to rhythm is a good music visualization design." **Average Score: 4.00**
- Q2: "The color changes of the sphere controlled by melody are a good music visualization design." **Average Score: 3.36**
- Q3: "Happiness and surprise are common emotions experienced during concerts." **Average Score: 4.50**
- Q4: "In the fantasy-style planetary scene, changing the number of satellites around the sphere based on audience count is a good visualization design." **Average Score: 4.13**
- Q5: "Changing the water surface based on audience emotions in both stage and fantasy scenes is an effective visualization design." **Average Score: 3.59**
- Q6: "Mapping VR controllers to virtual light sticks and interacting with the live performance based on rhythm is an appropriate interaction design." **Average Score: 4.32**

Overall, the average responses were positive, indicating that most participants found the system's design and functionalities to be feasible and effective. Visualization features, such as the deformation of the sphere and color changes synchronized with music (Q1, Q2), helped participants feel more emotionally engaged. Based on Q3, we will use an emotional model including "Natural," "Happy," and "Surprise" to better fit the concert setting. Visualizing onsite audience numbers and emotions (Q4, Q5) provided online audiences with a real-time visual representation of onsite presence and emotional states, strengthening social connectedness between online and onsite audiences.

4 LIMITATION AND FUTURE WORK

Our system is currently in its experimental phase and has several limitations that need to be addressed. Firstly, the current model's inability to precisely detect the performer's hand movements poses a significant challenge. Secondly, the ever-changing stage lighting in live performances can adversely affect the accuracy of skeletal detection models. To address this, we plan to incorporate depth cameras, which are less influenced by lighting conditions and can provide more reliable data for skeletal tracking. Future studies will aim to include more performers to gather comprehensive feedback on this critical aspect of the system.

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